

ENGINEERING CALCULATION NOTE / RETROSPECTIVE

Spindle vibration reduction

A hand model of the motor-end overhang

Prepared for Jerich Lee

Units: N, m, kg, s (dimensions also shown in mm)

Main result under the declared assumptions. A five-section, laterally clamped overhang gives a first bending estimate of **336.8 Hz by Rayleigh’s method** and **333.7 Hz with two retained coordinates**. The latter is **8.64% above** the report’s 307.17 Hz ANSYS result. The model captures a tail-dominated bending mechanism, but its image-estimated dimensions and idealized support prevent this from being a validated prediction of the real spindle.

The simplest useful calculation

$$\text{geometry + restraint} \longrightarrow k_{\text{eff}}, m_{\text{eff}} \longrightarrow f_1 \simeq \frac{1}{2\pi} \sqrt{\frac{k_{\text{eff}}}{m_{\text{eff}}}}$$

(1)

Rather than put the entire spindle mass on one spring, retain the changing bending stiffness and the fact that material near a nearly stationary support contributes little to a tail-dominated mode.

Model / evidence	First frequency	Versus 307.17 Hz
Uniform 60 mm steel cantilever, $L = 410$ mm	255.2 Hz	−16.93%
Stepped overhang, one Rayleigh coordinate	336.8 Hz	+9.64%
Stepped overhang, two-coordinate reduction	333.7 Hz	+8.64%
Original ANSYS result, Figure 18 / Table 5 ^[1]	307.17 Hz	Reference

What is retained, and what is deliberately left out

The local model retains the motor-side bearing location, overhang, large drive gear, stepped tail, report-based steel/aluminum properties, and distributed inertia. It omits the tool-side subassembly’s motion, housing flexibility, internal coaxial details, finite bearing stiffness, rotation, damping, and cutting loads. A support-flexibility extension is given on page 12.

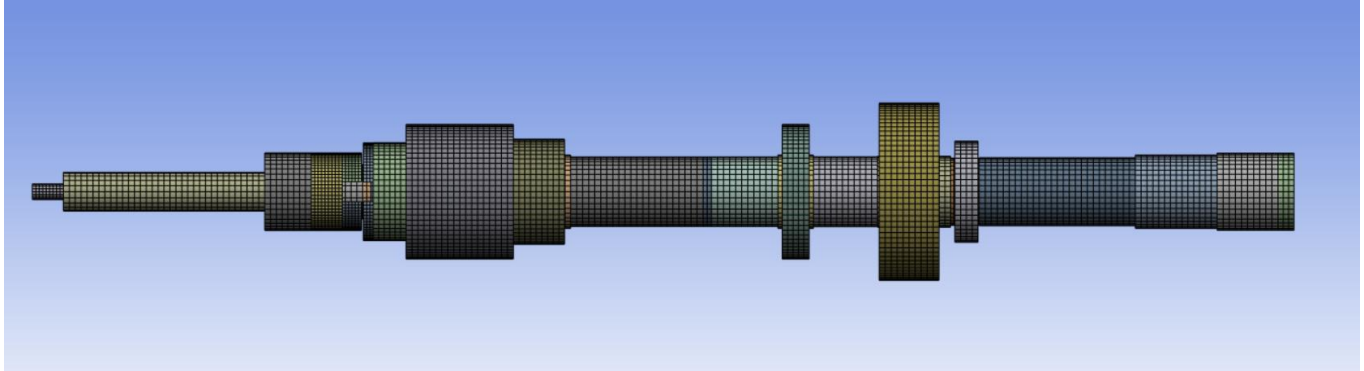
This is not a reconstruction of the original solver. No dimensioned spindle CAD, original ANSYS project, bearing stiffness data, or raw modal test data were supplied. Dimensions are rounded estimates from report figures; bores and internal interfaces are not resolved. No input was explicitly calibrated to 307 or 383 Hz. Nevertheless, this is a retrospective, mode-informed model, not a blind prediction.

Reading route: physical model and inputs, pp. 2–3; core hand calculation, pp. 4–8; ANSYS comparison and bearing change, pp. 9–10; sensitivities, interview explanation, and limitations, pp. 11–14; verification and references, pp. 15–16.

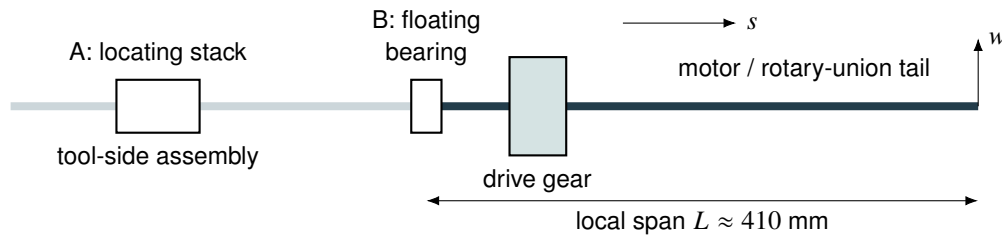
01 / PHYSICAL MODEL

Model the overhang that actually bends

The report's first ANSYS mode is concentrated at the motor-end tail, opposite the cutting tool. Its assembly has four locating bearings near the tool and a floating radial bearing closer to the drive gear. The rotary-union region extends beyond that floating bearing. ^[1, pp. 14–15, 31]



Original report Figure 13, PDF p. 28 (printed p. 27). Undeformed mesh; the thin flange before the large gear is the floating bearing. The smaller ring after the gear is not that bearing.



Why a lateral clamp is used at B

The report uses grounded outer bearing faces and *deformable cylindrical kinematic joints* at the shaft interfaces. ^[1, pp. 27–29] A cylindrical joint permits axial translation and axial spin, but restrains the two transverse translations and two transverse rotations. ^[3] Thus an ideal radial pin is not automatically equivalent to the recorded FE joint.

For this local reduction, take the *limiting* restraint

$$w(0) = 0, \quad w'(0) = 0. \quad (2)$$

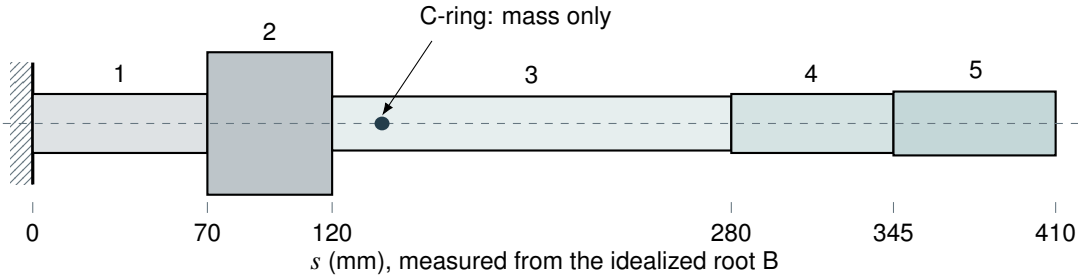
This condenses the tool-side assembly and bearing attachment into a rigid lateral root. It is not an exact replacement for the three-dimensional deformable coupling, bearing-body deformation, or actual rolling contacts. In ANSYS, “deformable” describes how the attachment geometry can deform; it does not itself supply a measured rolling-bearing stiffness. ^[4]

Spin and bending slope are different rotations. Restraining $w'(0)$ does not mean welding the spindle against rotation about its own axis. Likewise, “floating” means axial freedom in this assembly, not absence of radial support. Axial motion and spin are outside the present one-plane bending model.

The source report uses several global-axis conventions in its illustrations. Here s always means distance along the local shaft axis and w is one transverse displacement.

02 / INPUTS

A five-section approximation, not a CAD audit



Region	Assumed equivalent body	s (mm)	D (mm)	E (GPa)	ρ (kg/m ³)
1	Steel neck	0–70	60	205	7850
2	Bonded drive-gear envelope	70–120	145	210	7800
3	Aluminum tail stem	120–280	55	71	2770
4	Aluminum rear step	280–345	60	71	2770
5	Aluminum terminal step	345–410	65	71	2770

E and ρ : report Table 2. The report assigns steel to the shaft and gear, and aluminum to the rotary union/drawbar. Mapping the visible tail to regions 3–5, and treating the gear as one bonded section, are analyst assumptions rather than verified CAD assignments.^[1, pp. 25–27]

How the dimensions were estimated

Figure 18's 400 mm scale bar spans approximately 502 pixels in the extracted image. Matching the gear width between Figures 18 and 13 transfers an approximate scale of 1.5 pixels/mm to the undeformed mesh image. There, B is near pixel 976 and the tail end near 1590: $(1590 - 976)/1.5 \approx 409$ mm. The gear edges and tail shoulders give rounded stations near 70, 120, 280, and 345 mm. Diameter estimates come from the *undeformed* silhouette, not the exaggerated modal deflection.^[1, Figs. 13, 18]

Section and accessory assumptions

Use solid equivalent sections, $d = 0$:

$$A = \frac{\pi D^2}{4}, \quad I = \frac{\pi D^4}{64}, \quad \mu = \rho A. \quad (3)$$

Represent the thin C-ring by extra mass at $s_r = 140$ mm, with assumed OD/ID/width of 80/55/20 mm:

$$m_r = 7800 \frac{\pi}{4} (0.080^2 - 0.055^2) (0.020) = 0.4135 \text{ kg}. \quad (4)$$

Visible envelopes do not reveal the internal load path. No bore dimensions, drawbar clearances, bearing contact stiffnesses, or rotary-union interface stiffnesses are established. The solid gear includes its core once; the C-ring adds only annular mass. Coaxial internals are not separately added. These assumptions affect both stiffness and inertia and do not produce a rigorous bound.

03 / FIRST PASS

Start with the cantilever you can derive

The model you may be thinking of is the **Jeffcott rotor**: a simple shaft-and-disk idealization. Its familiar centered-disk form is useful for whirl and imbalance, but does not directly preserve this assembly's long, stepped overhang and distributed tail motion.^[6] Here the simplest starting point is a uniform Euler–Bernoulli cantilever, followed by a stepped refinement.

1. Write the equation and boundary conditions

For a uniform beam without rotation, damping, or applied load,

$$EI \frac{\partial^4 w}{\partial s^4} + \rho A \frac{\partial^2 w}{\partial t^2} = 0. \quad (5)$$

With $w(s, t) = Y(s) \cos \omega t$, the root is laterally clamped and the tail is free:

$$Y(0) = Y'(0) = 0, \quad Y''(L) = Y'''(L) = 0. \quad (6)$$

The free-end conditions are zero bending moment and zero shear force. Substitution gives

$$Y'''' - \left(\frac{\beta}{L}\right)^4 Y = 0, \quad \cosh \beta \cos \beta = -1. \quad (7)$$

Its first roots are $\beta_1 = 1.87510$ and $\beta_2 = 4.69409$, so

$$f_n = \frac{\beta_n^2}{2\pi L^2} \sqrt{\frac{EI}{\rho A}}. \quad (8)$$

2. Substitute a 60 mm steel envelope

For $L = 0.410$ m, $D = 0.060$ m, $E = 205 \times 10^9$ Pa, and $\rho = 7850$ kg/m³,

$$A = 2.82743 \times 10^{-3} \text{ m}^2, \quad I = 6.36173 \times 10^{-7} \text{ m}^4, \quad (9)$$

$$f_1 = \frac{1.87510^2}{2\pi(0.410)^2} \sqrt{\frac{(205 \times 10^9)(6.36173 \times 10^{-7})}{7850(2.82743 \times 10^{-3})}} = \boxed{255.2 \text{ Hz}}, \quad (10)$$

$$f_2 = \boxed{1599.1 \text{ Hz}}. \quad (11)$$

These are modes of the *uniform surrogate*, not identified modes of the complete spindle. Its first estimate is 16.93% below 307.17 Hz.

3. Learn the design scaling before adding detail

Because $I/A = D^2/16$ for a solid circle,

$$f_n \propto \frac{D}{L^2} \sqrt{\frac{E}{\rho}}. \quad (12)$$

Shorter overhangs and larger diameters tend to raise bending frequencies. A stronger steel grade does not necessarily have a much larger E/ρ . This helps explain why changing steel grades had little effect in the original report.^[1, Table 5]

Why refine this result? The stepped approximation has a wide, stiff gear near the root and a lighter aluminum tail with several diameter changes. One average diameter and one material cannot retain that distribution. The next calculation changes the distribution explicitly, rather than tuning the uniform beam to 307 Hz.

04 / STATIC FLEXIBILITY

Turn the stepped overhang into a spring

Apply a small conceptual transverse force P at $s = L$. This is a mathematical probe to obtain flexibility, not a claimed milling load. For the cantilever,

$$M(s) = P(L - s), \quad U = \frac{1}{2} \int_0^L \frac{P^2(L - s)^2}{E(s)I(s)} ds. \quad (13)$$

From $\delta_L = \partial U / \partial P$, the tail compliance and equivalent stiffness are

$$C_{LL} = \frac{\delta_L}{P} = \sum_j \frac{(L - a_j)^3 - (L - b_j)^3}{3E_j I_j}, \quad k_L = C_{LL}^{-1}. \quad (14)$$

Here a_j, b_j are the section endpoints in *meters*. Different sections have the same moment diagram but different curvatures $M/(EI)$.

A complete worked term: the 60 mm root neck

$$I_1 = \frac{\pi(0.060)^4}{64} = 6.36173 \times 10^{-7} \text{ m}^4, \quad (15)$$

$$C_1 = \frac{(0.410 - 0)^3 - (0.410 - 0.070)^3}{3(205 \times 10^9)(6.36173 \times 10^{-7})} = 7.56992 \times 10^{-8} \text{ m/N}. \quad (16)$$

Region	Body	$I \text{ (mm}^4\text{)}$	$EI \text{ (N m}^2\text{)}$	$C_j \text{ (10}^{-7} \text{ m/N)}$
1	Steel neck	636,173	130,415	0.75699
2	Gear envelope	21,699,109	4,556,813	0.01091
3	Tail stem	449,180	31,892	2.31951
4	Rear step	636,173	45,168	0.14187
5	Terminal step	876,241	62,213	0.01471
Total				3.24399

$$C_{LL} = 3.24399 \times 10^{-7} \text{ m/N}, \quad k_L = 3.08262 \times 10^6 \text{ N/m}. \quad (17)$$

As a units check, an illustrative $P = 100 \text{ N}$ at this motor-end tail gives

$$\delta_L = PC_{LL} = 3.24399 \times 10^{-5} \text{ m} = 32.44 \text{ } \mu\text{m}. \quad (18)$$

That is not the cutting-tool displacement: the force is applied at a different location, and the tool-side geometry is absent.

The weak span matters more than the largest visible part. The 160 mm aluminum stem contributes about 71.5% of this static bending compliance; the steel root neck contributes 23.3%. The wide gear contributes only 0.34%. These percentages describe this assumed beam, not extracted ANSYS strain-energy fractions.

05 / RAYLEIGH ESTIMATE

Use the mass that participates in the mode

Approximate the first mode by the static tip-load deflection $v(s)$ and normalize it as $\phi(s) = v(s)/v(L)$. This is a Rayleigh energy approximation; no vibration amplitude needs to be assumed.^[7]

Build the shape one section at a time

Within section j , let $u = s - a_j$. Integrating $v'' = P(L - s)/(E_j I_j)$ gives

$$v(s) = v(a_j) + v'(a_j)u + \frac{P}{E_j I_j} \left[\frac{(L - a_j)u^2}{2} - \frac{u^3}{6} \right], \quad (19)$$

$$v'(s) = v'(a_j) + \frac{P}{E_j I_j} \left[(L - a_j)u - \frac{u^2}{2} \right]. \quad (20)$$

Start from $v(0) = v'(0) = 0$ and carry the endpoint displacement and slope into the next section. The resulting shape is continuous in displacement and slope; curvature can change where EI changes.

Reduce the kinetic and strain energies

Set $w(s, t) = \phi(s)q(t)$ with $\phi(L) = 1$. Then

$$T = \frac{1}{2} \left[\underbrace{\sum_j \rho_j A_j \int_{a_j}^{b_j} \phi^2 ds + m_r \phi(s_r)^2}_{m_{\text{eff}}} \right] \dot{q}^2, \quad (21)$$

$$U = \frac{1}{2} k_L q^2, \quad m_{\text{eff}} \ddot{q} + k_L q = 0. \quad (22)$$

Region	Body	Mass (kg)	ϕ at right end	Modal mass (kg)
1	Steel neck	1.55367	0.02239	0.000159
2	Gear envelope	6.44007	0.05369	0.009823
3	Tail stem	1.05297	0.44748	0.059705
4	Rear step	0.50908	0.71522	0.173719
5	Terminal step	0.59746	1.00000	0.442895
–	C-ring extra mass	0.41351	–	0.002131
Total		10.56676		0.688433

Modal mass uses the integral of ϕ^2 , not each section's endpoint value times its mass. The shape is cubic within each material interval; its squared polynomial can be integrated directly.

$$f_R = \frac{1}{2\pi} \sqrt{\frac{3.08262 \times 10^6}{0.688433}} = 336.8 \text{ Hz.} \quad (23)$$

A large component need not dominate a mode's inertia. The 6.44 kg gear moves little in this shape and contributes only about 0.0098 kg to tip-normalized modal mass. The 0.60 kg terminal step contributes about 0.44 kg. Using all 10.57 kg as a tip mass would describe the wrong motion.

06 / TWO-COORDINATE MODEL

Obtain two modes without a large FE model

Retain two physical displacements: gear-center motion $q_1 = w(g)$ at $g = 0.095$ m and tail motion $q_2 = w(L)$ at $L = 0.410$ m. This is a **two-coordinate Rayleigh–Ritz reduction**, not two guessed lumped masses.

Static flexibility provides the interpolation

The displacement at s due to a unit transverse force at another axial station ξ is

$$H(s, \xi) = \int_0^{\min(s, \xi)} \frac{(s-u)(\xi-u)}{E(u)I(u)} du. \quad (24)$$

Each section contributes, with $z = \min(s, \xi, b_j) > a_j$,

$$H_j(s, \xi) = \frac{s\xi(z - a_j) - (s + \xi)(z^2 - a_j^2)/2 + (z^3 - a_j^3)/3}{E_j I_j}. \quad (25)$$

Let $s_1 = g, s_2 = L$ and $C_{ij} = H(s_i, s_j)$. Under forces $\mathbf{P}$, $\mathbf{q} = \mathbf{C}\mathbf{P}$. Hence the reduced stiffness is $\mathbf{K} = \mathbf{C}^{-1}$, and the reconstructed beam shape is

$$w(s, t) = \mathbf{N}(s)\mathbf{q}(t), \quad \mathbf{N}(s) = \begin{bmatrix} H(s, g) & H(s, L) \end{bmatrix} \mathbf{C}^{-1}. \quad (26)$$

The interpolation exactly returns $\mathbf{N}(g) = [1 \ 0]$ and $\mathbf{N}(L) = [0 \ 1]$.

Keep the continuous mass distribution

$$\mathbf{M} = \sum_j \rho_j A_j \int_{a_j}^{b_j} \mathbf{N}^\top \mathbf{N} ds + m_r \mathbf{N}(s_r)^\top \mathbf{N}(s_r). \quad (27)$$

The entries follow by integrating products of piecewise cubic functions, just as on the previous page. Stiffness can also be checked from $\int E I N'' T N'' ds$.

$$\mathbf{C} = 10^{-7} \begin{bmatrix} 0.0215260 & 0.1231871 \\ 0.1231871 & 3.2439936 \end{bmatrix} \text{ m/N}, \quad (28)$$

$$\mathbf{K} = 10^6 \begin{bmatrix} 593.53793 & -22.53896 \\ -22.53896 & 3.938513 \end{bmatrix} \text{ N/m}, \quad (29)$$

$$\mathbf{M} = \begin{bmatrix} 12.188384 & 0.788953 \\ 0.788953 & 0.610937 \end{bmatrix} \text{ kg}. \quad (30)$$

M_{11} is not the physical gear mass, nor is M_{22} the physical tail mass. They are generalized inertia coefficients for the selected displacement coordinates. The off-diagonal term is retained because the interpolated motions share distributed material.

Rayleigh–Ritz trial shapes must satisfy the root kinematic constraints. They need not satisfy the unloaded free-end shear condition exactly: natural boundary conditions are approached through the energy approximation as the trial space is enriched.

07 / NATURAL FREQUENCIES AND SHAPES

Solve a quadratic, then inspect the motion

Free vibration satisfies $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{0}$. With $\mathbf{q} = \mathbf{a} \cos \omega t$,

$$\det(\mathbf{K} - \lambda \mathbf{M}) = 0, \quad \lambda = \omega^2. \quad (31)$$

For symmetric 2×2 matrices this becomes

$$0 = (M_{11}M_{22} - M_{12}^2)\lambda^2 - (K_{11}M_{22} + K_{22}M_{11} - 2K_{12}M_{12})\lambda + (K_{11}K_{22} - K_{12}^2). \quad (32)$$

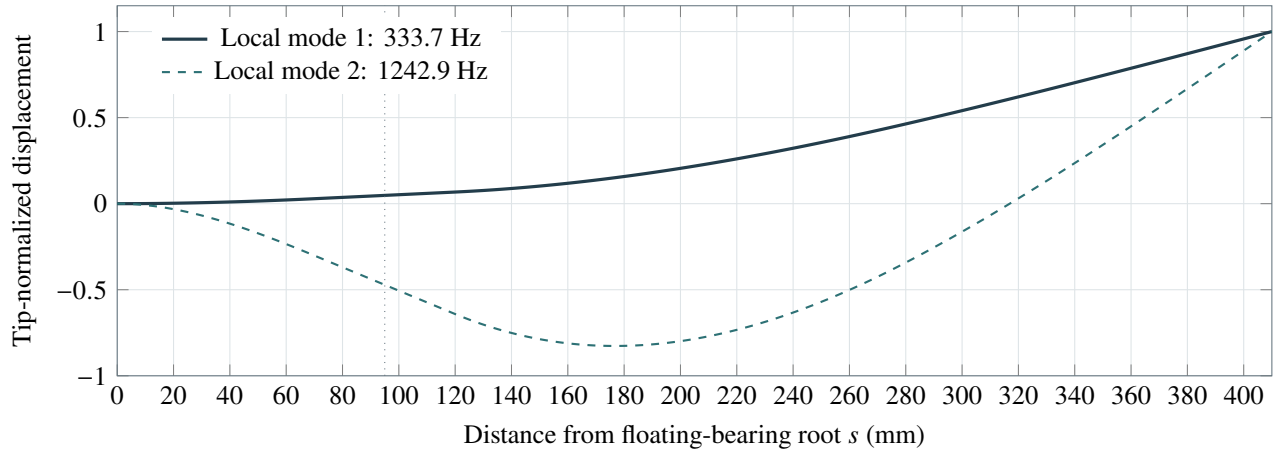
Using $x = \lambda/(10^6 \text{ s}^{-2})$ and dividing by a common dimensional factor gives the numerical quadratic

$$6.823892x^2 - 446.182993x + 1829.651991 = 0. \quad (33)$$

Thus $x_1 = 4.396264$, $x_2 = 60.989148$, and

$$f_{1,\text{local}} = 333.7 \text{ Hz}, \quad f_{2,\text{local}} = 1242.9 \text{ Hz}. \quad (34)$$

For unit tail displacement, the coordinate vectors are $\mathbf{a}_1 = [0.04817 \ 1]^\top$ and $\mathbf{a}_2 = [-0.47161 \ 1]^\top$. Reconstruct each shape from $\psi_n(s) = \mathbf{N}(s)\mathbf{a}_n$.

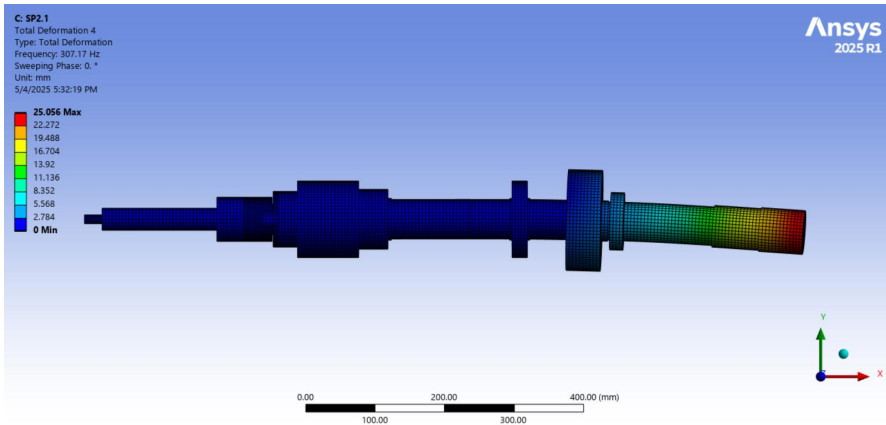


Computed shapes of the assumed overhang, not measured or digitized ANSYS eigenvectors. The dotted line marks the gear center. Mode 2 has an interior zero near $s = 317$ mm and opposing gear/tail motion.

The second local mode is not the report's second system mode. Table 5 lists 523.67 Hz, but supplies no corresponding mode shape. The omitted tool, inter-bearing span, joints, or another motion family can produce modes below 1243 Hz. No matched-mode error is assigned to that pair. In the nonrotating circular idealization, each local bending family also has an equal-frequency partner in the orthogonal plane.

08 / COMPARISON WITH THE ORIGINAL MODEL

Agreement in scale, not a validation claim



Original Figure 18, PDF p. 32 (printed p. 31), unmodified. The screenshot reads 307.17 Hz. Its maximum deformation legend is eigenvector scaling, not measured vibration amplitude.^[1,5]

Define the comparison before calling it an “error”

Use the FE readout as a reference, not as experimental truth:

$$e_{\text{HC|FE}} = 100 \frac{f_{\text{HC}} - f_{\text{FE}}}{f_{\text{FE}}}.$$

(35)

$$e_R = 100 \frac{336.7824 - 307.17}{307.17} = +9.64\%,$$

(36)

$$e_{2q} = 100 \frac{333.7045 - 307.17}{307.17} = \boxed{+8.64\%}.$$

(37)

The two-coordinate result is 26.53 Hz higher. Its first shape is qualitatively tail-dominated, like the shown FE mode. The report does not provide nodal eigenvectors for a quantitative shape correlation such as a modal assurance criterion.

Keep three distinct checks separate

Check	What the supplied record supports
Hand model versus FE	About 9% frequency discrepancy, but geometry, support representation, rotational treatment, and retained subassembly differ.
FE mesh sensitivity	Reported first frequencies of 305–307 Hz across four meshes. This is mesh sensitivity, not a physical error bound. ^[1, Table 3]
FE versus observation	An operational band near 310 Hz; 307.17 Hz is 0.91% lower. Impact tests were inconclusive, so a measured full mode shape is not established by the report. ^[1, pp. 19, 22, 30]

The report calls the model “validated.” Here the evidence supports **frequency correlation**, not a measured full mode shape or proof of loaded cutting stability. The new hand calculation is a separate, assumption-dependent comparison.

09 / BEARING MODIFICATION

Separate a support shift from a new bearing stack

The report associates the recommended modification with moving the floating stack 22 mm toward the drive and a first-mode shift from approximately 307 to 383 Hz. Figure 21 reads 383.27 Hz.^[1, pp. 33–34]

A controlled hand-calculation sensitivity

Move only the idealized root by $\Delta = 22$ mm, leaving the physical gear and tail geometry unchanged. In the new coordinate $s' = s - \Delta$,

$$L' = 0.410 - 0.022 = 0.388 \text{ m}, \quad g' = 0.073 \text{ m}. \quad (38)$$

The first section shortens from 70 to 48 mm. The remaining section lengths do not change; the C-ring moves to $s'_r = 118$ mm in the new coordinate. Repeating exactly the same calculations gives

Quantity	Original root	Root shifted 22 mm
Tip compliance (10^{-7} m/N)	3.24399	2.97537
Tip stiffness (10^6 N/m)	3.08262	3.36093
Rayleigh effective mass (kg)	0.68843	0.65662
Rayleigh frequency (Hz)	336.78	360.07
Two-coordinate first frequency (Hz)	333.70	358.30

$$\frac{358.3014}{333.7045} - 1 = 7.37\% \quad \text{versus} \quad \frac{383.27}{307.17} - 1 = 24.77\% \quad (\text{reported FE}). \quad (39)$$

What this comparison says

The translation-only model predicts the expected upward trend, but does *not* reproduce the reported 24.8% gain. Its already-clamped root leaves no additional root tilt to suppress. Adding and spacing real bearings, or changing FE joint scopes, can alter effective restraint beyond merely shifting one point. Their contribution cannot be quantified from the available geometry and setup.

Although 358.30 Hz is only 6.51% below 383.27 Hz, that scalar proximity is **not** agreement for the actual redesign: the hand case and FE modification are different cases.

A useful inverse check, explicitly not a prediction

For unchanged effective mass, the reported frequency ratio would require

$$\frac{k_{\text{new}}}{k_{\text{old}}} = \left(\frac{383.27}{307.17} \right)^2 = 1.55687, \quad (40)$$

or about 55.7% more effective stiffness. This is inferred from the frequencies, not independently calculated bearing stiffness.

The bearing count is inconsistent within the report. Section 7.8 and the synopsis describe adding one bearing (two total); the following page and Figure 21 say two additional bearings. Table 4 labels the 383 Hz case “Modified 2x.” This note preserves that label and does not silently decide which physical stack was simulated. Confirm the original CAD/joint scopes before reconstructing it.

10 / SENSITIVITY

The input uncertainty is larger than the arithmetic

The decimal places in the calculation allow reproduction; they are not dimensional accuracy or a statistical confidence interval. The image estimates, internal geometry, and support model deserve more scrutiny than the last hertz.

Geometric scaling within this idealized model

With unchanged materials, scaling all diameters by γ gives $EI \propto \gamma^4$, mass per length $\propto \gamma^2$, and $f \propto \gamma$. Scale the C-ring’s OD and ID with the other diameters. Scaling every axial length by α (including ring width and station positions) gives $f \propto \alpha^{-2}$. Scaling *all dimensions* by α gives $f \propto \alpha^{-1}$.

Assumed change from the baseline	First mode (Hz)	Change
All axial lengths −5%; diameters fixed	369.8	+10.80%
All axial lengths +5%; diameters fixed	302.7	−9.30%
All diameters −5%; axial lengths fixed	317.0	−5.00%
All diameters +5%; axial lengths fixed	350.4	+5.00%
All dimensions −5% together	351.3	+5.26%
All dimensions +5% together	317.8	−4.76%

These are chosen perturbations, not measured image-error bounds. A common image-scale error changes lengths and diameters together; it is not the same scenario as changing axial lengths alone.

Hollow and coaxial regions need both stiffness and mass changes

For an actual tube,

$$A = \frac{\pi}{4}(D^2 - d^2), \quad I = \frac{\pi}{64}(D^4 - d^4), \quad \frac{I}{A} = \frac{D^2 + d^2}{16}.$$

(41)

Removing material lowers both EI and mass. A uniform tube with the same outer diameter can have a *higher* self-mass bending frequency than a solid beam, even though its static stiffness is lower. Therefore “assuming solid must overpredict frequency” is not generally valid. A separate tip mass or coaxial contact arrangement changes the conclusion again.

Refinement order

First replace the image geometry with section dimensions and component masses from CAD. Next verify the effective bending restraint and the rotary-union/shaft connection. Then add shear deformation, rotary inertia, and speed-dependent terms as needed. The equivalent $L/D \approx 6.8$ and the short, wide gear are reasons not to treat Euler–Bernoulli higher modes as high-fidelity predictions.

Do not tune the length until the answer becomes 307 Hz. Keep independently obtained geometry fixed, calculate the discrepancy, then test physical hypotheses such as support compliance or a different internal load path. Matching one frequency cannot uniquely identify all of these inputs.

11 / BOUNDARY-CONDITION EXTENSION

What changes when the bearing can move or tilt?

For a real assembly, replace the ideal root with a translational spring k_y and bending-rotation spring k_θ . Assume these springs are massless, linear, uncoupled, and referred to ground. The tip compliance becomes

$$C_{LL,\text{total}} = C_{LL,\text{beam}} + \frac{1}{k_y} + \frac{L^2}{k_\theta}. \quad (42)$$

A force P causes root translation P/k_y and moment PL , giving slope PL/k_θ and extra tip motion PL^2/k_θ .

The same two-coordinate method still works

Replace the influence function on page 7 by

$$H_{\text{total}}(s, \xi) = H_{\text{beam}}(s, \xi) + \frac{1}{k_y} + \frac{s\xi}{k_\theta}. \quad (43)$$

Rebuild $\mathbf{C}$, $\mathbf{N}$, and $\mathbf{M}$, not just $\mathbf{K}$: support motion changes the modal mass distribution too. A more general root compliance matrix can include translation–rotation coupling. No numerical bearing stiffness is assigned here because none is documented.

Why two radial pins are not a clamp

For a *separate uniform-beam example*, place radial supports at $A = -a$ and $B = 0$, with a tail load P at L . Static equilibrium gives

$$R_A = -\frac{PL}{a}, \quad R_B = P \left(1 + \frac{L}{a}\right). \quad (44)$$

With constant EI , free tilt at each support, and radial stiffnesses k_A, k_B , energy in both spans and supports gives

$$C = \frac{L^2(a+L)}{3EI} + \frac{(L/a)^2}{k_A} + \frac{(1+L/a)^2}{k_B}. \quad (45)$$

Even for $k_A, k_B \rightarrow \infty$, the inter-bearing span supplies extra flexibility $L^2a/(3EI)$ compared with a local clamp. This is a different restraint from the primary calculation; it is not an extra correction to add blindly to an already condensed assembly.

How spacing a bearing pair can resist tilt

For two identical radial springs, each stiffness k , separated axially by d , their displacements under a small rigid-section translation y and tilt θ are $y \mp d\theta/2$. Therefore

$$U = \frac{1}{2}k(y - d\theta/2)^2 + \frac{1}{2}k(y + d\theta/2)^2 \quad (46)$$

$$= \frac{1}{2}(2k)y^2 + \frac{1}{2}\left(\frac{kd^2}{2}\right)\theta^2, \quad k_y = 2k, \quad k_\theta = \frac{kd^2}{2}. \quad (47)$$

Bearing spacing can matter as much as bearing count. A pair supplies a moment-resisting lever arm, while a single ideal radial point spring does not. Axial thermal freedom must still be preserved in the actual arrangement. This mechanism is explanatory; it is not a reconstruction of the ambiguous “Modified 2x” FE case.

12 / INTERVIEW EXPLANATION

The story to explain at a whiteboard

A concise, historically accurate explanation

“The original ANSYS mode showed the motor-end overhang bending much more than the tool-side assembly. I later built a hand model of that region to understand the result. I treated the bearing attachment as an ideal lateral clamp for the first pass, then represented the overhang as a stepped beam with the gear and aluminum tail.

I applied a unit transverse force to obtain its flexibility. That gave an effective stiffness of about 3.08 MN/m. I used the deflected shape to weight the distributed mass, rather than putting the full assembly mass at the end. The resulting effective mass was about 0.688 kg, giving about 337 Hz. A two-coordinate version gave 334 Hz, roughly 8.6 percent above the reported 307 Hz.

The calculation captures the bending scale, but I would not claim it validates the real spindle: the dimensions were estimated from report figures, and the bearing and internal interfaces need verification. Moving the support closer to the tail raises the frequency; a second axially spaced bearing can also resist tilt. My translation-only model did not reproduce the whole reported redesign gain.”

Why did you choose a cantilever rather than simple supports?

The dominant motion is beyond the motor-side bearing. Also, the recorded cylindrical-joint idealization restrains transverse rotations as well as translations. I used a local-clamp limit to reflect that constraint, not because a real rolling bearing is automatically a perfect clamp. The two-radial-support alternative is shown separately on page 12.

Why is the effective mass so much smaller than the total mass?

Kinetic energy weights mass by the square of its modal displacement. A heavy gear near a nearly stationary support can contribute less than a much lighter tail moving through the full amplitude. “Effective mass” here means tip-normalized generalized mass, not the ANSYS base-excitation participation metric.

What is your second mode, and why does it differ from ANSYS?

It is the second bending mode of the *retained overhang*, around 1243 Hz, with opposite gear and tail motion. The original report lists a second *system* frequency of 523.67 Hz without its shape. I cannot identify those as the same mode, because my model deliberately removes other parts of the system.

What did the hand calculation establish about the redesign?

It explains the direction of the support-location effect and the role of tilt restraint. A 22 mm translation alone raises my first mode by about 7.4%, not the report’s 24.8%. I would inspect the actual bearing count, spacing, joint scopes, and mode shapes before claiming a quantitative prediction of that redesign.

Preserve the chronology. The team project and its ANSYS results are from spring 2025. These hand calculations were added in September 2026. Do not describe them as analyses that drove the original design decision, and do not describe a simulated modification as a measured post-installation improvement without that evidence.

13 / DYNAMICS LIMITS

Frequency relocation is not vibration amplitude

Spindle speed and tooth-passing frequency are different

The report specifies up to 3000 rpm and six flutes.^[1, pp. 8–9] Therefore

$$f_{\text{spin,max}} = \frac{3000}{60} = 50 \text{ Hz}, \quad f_{\text{tooth,max}} = 6 \frac{3000}{60} = 300 \text{ Hz}. \quad (48)$$

The 307 Hz mode lies near the top of the *six-tooth fundamental* range, not near the once-per-revolution frequency at 3000 rpm. Equating a constant 307.17 Hz natural frequency with 1X would give about 18,430 rpm, outside that operating range. This conversion is not a rotordynamic critical-speed prediction.

Raising a mode above 300 Hz separates it from the maximum fundamental tooth-passing frequency, but does not remove higher harmonics, broadband excitation, or regenerative cutting instability. The unloaded local modal model does not establish chatter-free operation.

This calculation is nonrotating; the reported FE setup is not

The report records 3000 rpm, Coriolis effects, and a damped modal solver. It excludes the bearings and rotary union from the assigned rotation.^[1, p. 28] The present model omits those terms, so its comparison is not at matched speed and damping conditions. A general rotating linearized model has the form

$$\mathbf{M}\ddot{\mathbf{q}} + (\mathbf{C}_d + \Omega\mathbf{G})\dot{\mathbf{q}} + \mathbf{K}(\Omega)\mathbf{q} = \mathbf{f}(t). \quad (49)$$

Gyroscopic and prestress effects require the spinning-component inertias, interface constraints, and a properly configured speed-dependent analysis.^[5] The fact that $50/307 \approx 0.16$ is not by itself an error bound for omitting them. A matched zero-speed FE extraction would be the clean first comparison.

A modal displacement legend does not predict operating motion

In an eigenproblem, any multiple of an eigenvector is still an eigenvector. The 25.056 mm maximum shown in the original first-mode image is therefore not a physical vibration amplitude. ANSYS likewise cautions that modal displacement/stress magnitudes are not actual response values.^[5]

For a single forced mode,

$$|q(\omega)| = \frac{|F_{\text{modal}}|/k_{\text{modal}}}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}, \quad r = \frac{\omega}{\omega_n}. \quad (50)$$

Amplitude needs the forcing, damping, and relevant mode shape. A frequency shift alone does not demonstrate a reduction in measured RMS velocity, meet an ISO amplitude limit, or prove increased damping.

Keep unloaded and cutting claims separate. The original report states that its loaded simulation was not successfully validated. Its impact tests were inconclusive, and some loaded long-tool recordings saturated.^[1, pp. 19, 24–25] This hand note does not repair those evidence gaps or derive the nonlinear cutting response from an unloaded natural frequency.

14 / REPRODUCIBILITY

Check the reduction separately from the real machine

An internal numerical check of the same assumed beam

A small, independent Euler–Bernoulli beam discretization with consistent distributed mass was assembled in Python. It uses the same five sections, same ring mass, and same clamped root. This is a verification of the reduced calculation, **not a new ANSYS analysis or physical validation**.

Discretization / approximation	Local mode 1 (Hz)	Local mode 2 (Hz)
6 beam elements	331.16	1228.14
12 beam elements	331.15	1227.25
24 beam elements	331.15	1227.20
One-coordinate Rayleigh approximation	336.78	Not retained
Two-coordinate Ritz approximation	333.70	1242.93

Relative to the 24-element check, the two-coordinate frequencies are 0.77% and 1.28% higher. The single-coordinate first estimate is 1.70% higher. The positive differences are consistent with restricted Rayleigh–Ritz trial spaces for this self-adjoint beam problem; they are not upper bounds relative to the actual assembly or original rotating FE model.

What the calculation script verifies

`verify_spindle.py` recomputes section stiffnesses, static flexibility, generalized masses, both eigenvalues, the 22 mm shift, sensitivities, and all comparison percentages. It checks symmetry, positive mass, station interpolation, stiffness from strain energy, eigenvector mass orthogonality, and eigen-equation residuals. Four-point Gaussian integration on each polynomial interval exactly integrates the mass terms in exact arithmetic; floating-point roundoff remains.

Editable package	Purpose
<code>spindle_hand_calculations.tex</code>	Typeset report and vector schematics.
<code>verify_spindle.py</code>	Standalone calculation and internal beam check; edit section inputs at the top.
<code>data/results.json</code>	Full-precision inputs, matrices, and numerical results.
<code>data/mode_shapes.dat</code>	Computed curves used by the LaTeX plot.
<code>assets/</code> and <code>README.md</code>	Original extracted figures, provenance, limitations, and rebuild instructions.

What would turn this into a stronger model-to-model check?

Replace the silhouette estimates with a CAD station table, including bores and coaxial parts. Record each component’s mass and connection path. Recover the baseline and modified joint locations/scopes, exact bearing count and spacing, and modal speed settings. Then compare corresponding zero-speed mode shapes as well as frequencies. Finite bearing stiffness and housing motion should be added only with identified parameters or explicitly labeled sensitivity cases.

Current conclusion: the hand calculation explains a plausible motor-end bending mechanism and lands within about 9% of the reported first frequency. Its mathematical reduction is much more tightly checked than its physical inputs. That distinction is the central engineering lesson of this retrospective exercise.

15 / REFERENCES AND EVIDENCE

Where the inputs and comparisons come from

[1] **Primary project record.** Z. Balakrishnan, R. VandeWiele, E. Mahmudi, L. Wydra, J. Lee, and J. Frank. *Spindle Vibration and Deflection Reduction*. University of Illinois Urbana–Champaign, Spring 2025 ME 470 final report, 9 May 2025. User-supplied file: SVRSP25FinalReport–2.pdf. PDF page numbers are one greater than the report’s printed numbers after the title page.

PDF pages	Specific evidence used in this note
10, 15–16	Speed/flutes; fixed and floating bearings; shaft, gear, drawbar, rotary union, C-ring anatomy.
20, 23, 25	Inconclusive impact tests; unloaded band near 310 Hz; saturated loaded data.
26–30	Material assignments and Table 2; cylinder approximation; Figure 13 mesh; joint, grounding, rotation, and solver descriptions.
31–32	Frequency tuning narrative; Figure 18 at 307.17 Hz; mesh results 305–307 Hz.
34–35, 42	22 mm relocation; inconsistent bearing counts; Figure 21 at 383.27 Hz; reported modification series.
37	Table 5: baseline 307.17 / 523.67 Hz and steel-grade comparisons.

[2] **Project portfolio.** J. Lee. *Spindle Vibration & Deflection Reduction*. Accessed 7 September 2026. [Project page](#). Used for project context; the more detailed supplied report controls numerical and setup interpretation. The page’s broader claims of mode-shape validation and safe separation from cutting excitation are not independently established by this note.

[3] **Joint degrees of freedom.** ANSYS. *Multibody Analysis Guide*, release 2024 R2, §2.3, “Connecting Multibody Components with Joint Elements.” [Official documentation](#). In particular, the cylindrical-joint constraint list. Consulted 7 September 2026; the original project deck was not available for an element-level audit.

[4] **Attachment behavior.** ANSYS. *Mechanical User’s Guide*, release 2024 R2, §12.2, “Geometry Behaviors.” [Official documentation](#). Deformable versus rigid remote geometry attachment. Consulted 7 September 2026.

[5] **Modal interpretation.** ANSYS. *Mechanical User’s Guide*, release 2025 R2, §5.6.5, “Modal Analysis.” [Official documentation](#). Modal scaling, joint constraints, damped/rotating analyses, and prestress discussion. Consulted 7 September 2026. Referencing these instructions does not confirm the original model’s actual solver implementation.

[6] **Simple rotor models.** J. E. T. Penny, M. I. Friswell, A. W. Lees, and S. D. Garvey. *A simple but versatile rotor model*. C623/033/2004, 2004, opening discussion on printed p. 269. [Author-hosted paper](#). Supports the Jeffcott-model context; the present stepped-overhang derivation is a separate reduction.

[7] **Rayleigh energy method.** University of Alberta, Engineering at Alberta Courses. *Vibrations and Sound: Application to Lateral Vibrations of Beams*. [University course resource](#). Method background; all numerical sections, matrices, and modal curves in this note were computed for the explicitly assumed geometry.

[8] **Methodological precedent.** *Pigjaw push-out fixture: Hand calculations, retrospective PLA FEA, and model comparison*, CALC-01 Rev. C, 7 September 2026. User-supplied file: *Pigjaw_Fixture_Material_Hand_Calculations_Rev_C.pdf*. Used as the model for separating sourced inputs, assumptions, model discrepancies, and retrospective project history; no spindle dimensions or properties are taken from it.

Evidence boundary. This package contains no original spindle CAD, solver archive, raw modal test records, measured bearing stiffness, or as-built verification. Extracted report figures are preserved as illustrations, not newly generated simulations. The calculation is an engineering explanation and screening model, not a load rating, commissioning assessment, or certification of machine performance.